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Determination of Work Function of Graphene under a Metal Electrode and Its Role in Contact Resistance

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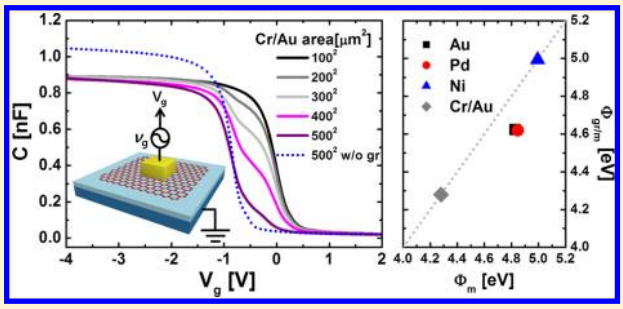
Supporting Information

ABSTRACT: Although the work function of graphene under a given metal electrode is critical information for the realization of high-performance graphene-based electronic devices, relatively little relevant research has been carried out to date. In this work, the work function values of graphene under various metals are accurately measured for the first time through a detailed analysis of the capacitance−voltage ($C-V$) characteristics of a metal−graphene−oxide−semiconductor (MGOS) capacitor structure. In contrast to the high work function of exposed graphene of 4.89−5.16 eV, the work function of graphene under a metal electrode varies depending on the metal species. With a Cr/Au or Ni contact, the work function of graphene is pinned to that of the contacted metal, whereas with a Pd or Au contact the work function assumes a value of ∼4.62 eV regardless of the work function of the contact metal. A study of the gate voltage dependence on the contact resistance shows that the latter case provides lower contact resistance.



KEYWORDS: *Graphene, work function, contact resistance, capacitance−voltage, flat band voltage*

In graphene electronics, the contact resistance (R_c) between graphene and metal limits the performance of the device, such as the carrier mobility and ON-state current of graphene field effect transistors (FETs).[1−10] Furthermore, the best contact resistance achieved thus far is still an order of magnitude higher than the R_c between a silicon source/drain and metal. To date, most studies on metal−graphene contact resistance have focused on the charge transport or the interface between graphene and metal.[1,2,8−16] Previous works reported that Pd or Ni provides a relatively low contact resistance, whereas higher contact resistance was observed with Ti, Cr, and Al contacts.[1,8,9,16,17] The gate bias has also been found to contribute to the R_c, by changing the charge density in the graphene channel.[1,2,5,9,12] It also has been reported that the R_c is strongly affected by the conduction mode of graphene under metal and the graphene channel, and the high work function of graphene (Φ_{gr}) leads to better carrier injection efficiency from metal to graphene.[9,10]

A Ti contact is meanwhile expected to have a smaller number of conduction modes in graphene and thereby provide higher contact resistance compared to a Pd contact.[9] However, the exact work function of graphene under Ti has not been clearly ascertained, and thus there is no evidence that the higher conduction modes in graphene under metal lead to lower contact resistance. Although progress has been made in clarifying relevant issues, several important questions remain unanswered: (1) Is graphene under metal still graphene? (2) Is the original work function of graphene preserved when graphene contacts metal? (3) Does high work function graphene under metal indeed provide low contact resistance? (4) Is the work function of graphene under metal the sole factor responsible for the low contact resistance? Better understanding related to these issues is essential for the realization of graphene-based electronic devices. A similar example can be found in advanced CMOS technology: research on the work function of the gate metal on high-κ dielectrics is critical in deep scaled CMOS devices.[18−20] In graphene-based electronic devices, graphene employed as a channel material or as an electrode material always contacts a metal, and thus the work function of graphene under metal ($\Phi_{gr/m}$) must be carefully studied.

It was theoretically predicted that graphene on a certain metal substrate such as Al, Au, or Pt could be doped by charge transfer due to the presence of an interface dipole layer between graphene and the metal.[21,22] It was also experimentally found that the metal contact induces charge inhomogeneity,[2,5,11] resulting in a difference between the work function of graphene under metal and that of bare graphene, as measured by Kelvin probe force microscopy (KPFM)[23,24] and scanning photocurrent microscopy.[25−28] This difference in the work function leads to band bending at the contact edge.[24−27] The reported values of $\Phi_{gr/m}$, however, are evaluated from indirect measurement, for example, by finding the gate voltage that yields a flat channel potential profile.[25−28] Despite the significance of the work function, only theoretical calculation results have been reported to date, and there is a lack of direct

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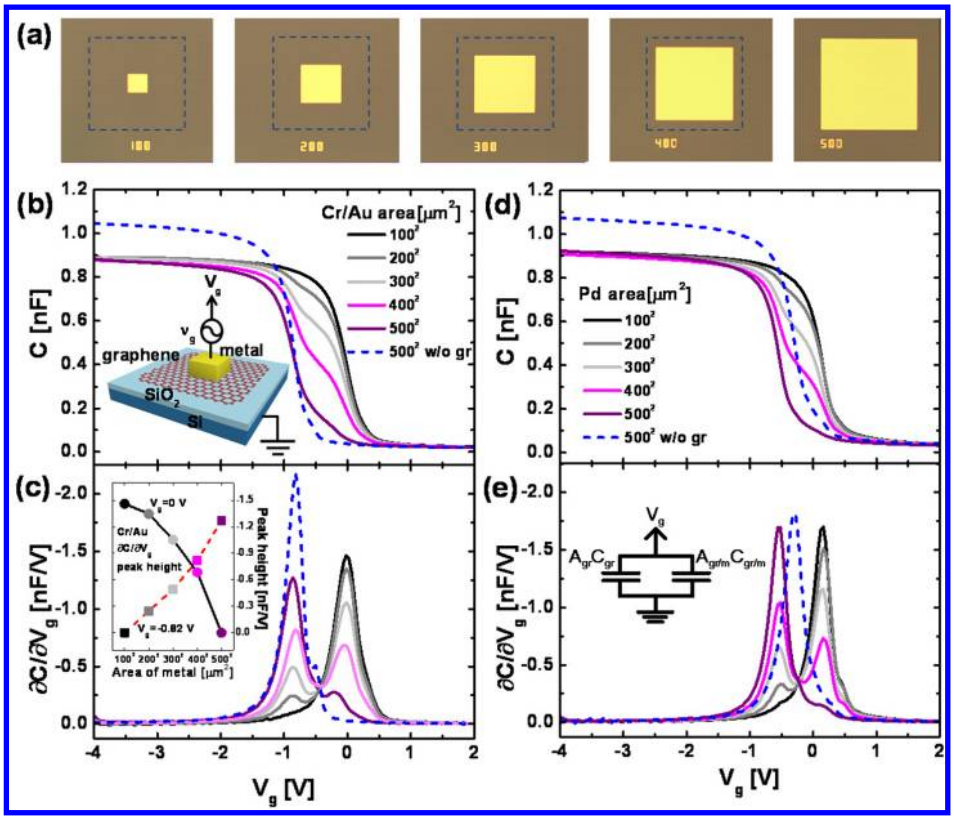

Figure 1. (a) Optical microscope images of MGOS capacitors. The dashed line indicates the area of graphene while yellow squares are the metal pad on graphene. (b) $C-V$ curves of MGOS capacitors for different coverage of Cr/Au on graphene. $C-V$ curve of MOS capacitor (Cr/Au electrode without graphene) is also plotted together with dashed line. The measurement condition with MGOS capacitor is illustrated in the inset. (c) Differentiation ($\partial C/\partial V_g$) of the $C-V$ curves in panel b. The inset shows the height of the peak value of $\partial C/\partial V_g$ at $V_g = -0.82$ and 0 V as a function of the metal area. (d) $C-V$ curves of MGOS and MOS capacitors but with a Pd metal gate instead of a Cr/Au gate. (e) $\partial C/\partial V_g$ of the $C-V$ curves in panel d. The inset illustrates a model of a MGOS capacitor with two parallel capacitors of $C_{gr/m}$ (capacitance by metal covered area) and C_{gr} (capacitance by exposed graphene area).

experimental observations on how the work function of graphene is affected by different metals. Here, we report new and important findings on the properties of graphene under metal, especially the behavior of the work function, obtained through a detailed and thorough analysis of capacitance–voltage ($C-V$) measurements using a metal–graphene–oxide–semiconductor (MGOS) capacitor structure. This is the first time this approach has been utilized. The $C-V$ measurement method is simple, accurate, and widely used to extract the work function of metal electrodes in silicon MOS device research and is applied here to a graphene device to study the work function of graphene under metal.

As the flat band voltage (V_{FB}) of a metal–oxide–semiconductor (MOS) capacitor is directly related to the work function of the gate material, $\Phi_{gr/m}$ can be obtained from $C-V$ measurements with good accuracy. First, large area monolayer graphene was grown on a thin Cu film by inductively coupled plasma chemical vapor deposition (ICP-CVD) and transferred onto a designated wafer by the conventional wet transfer method with PMMA.[29−33] The quality of the transferred graphene was confirmed by the low intensity of the Raman D band, as shown in Supporting Information, Figure S1. The thickness of the graphene was confirmed by atomic force microscopy (AFM) and transmittance measurements. The CVD-grown graphene was transferred to a thermally grown SiO_2 layer on a p-type (resistivity of ∼10 Ω·cm) silicon substrate. Metal pads with various areas were formed on the transferred graphene, followed by patterning of the graphene with an area of 500 × 500 μm^2, as shown in Figure 1a. In this experiment, various metals such as a Cr/Au stack, Ni, Au, and a Pd/Au stack were respectively used in order to study the work function behavior of graphene under different metal electrodes. The thickness of the metal was 5/100 nm for Cr/Au, 50 nm for Ni, 100 nm for Au, and 20/80 nm for Pd/Au. The metal pad patterns were defined by a lift-off process. The $C-V$ curves were measured in an air ambient at room temperature using an Agilent 4284A Precision LCR meter with a frequency of 1 kHz, as illustrated in the inset of Figure 1b. The contact resistances were measured using the transmission line method (TLM). A MOS capacitor structure without graphene was also prepared for comparison. It must be noted that graphene serves as a gate electrode in our test device structure, which is in contrast to previous graphene quantum capacitance studies where graphene serves as a silicon body in MOS devices,[34,35] and thus the purpose and the structure of the test device are comprehensively different.

The $C-V$ curve of the MOS capacitor without graphene (hereafter referred to as the "control sample", indicated by the blue dotted curve in Figure 1b) shows a typical well-behaved $C-V$ relationship for a MOS capacitor (accumulation–depletion–inversion) with a V_{FB} of ∼−0.82 V. However, the $C-V$ curve obtained from the MGOS structure with a Cr/Au area of 100 × 100 μm^2 (black solid line) shows a positive shift relative to the control sample and the V_{FB} is ∼0 V. An

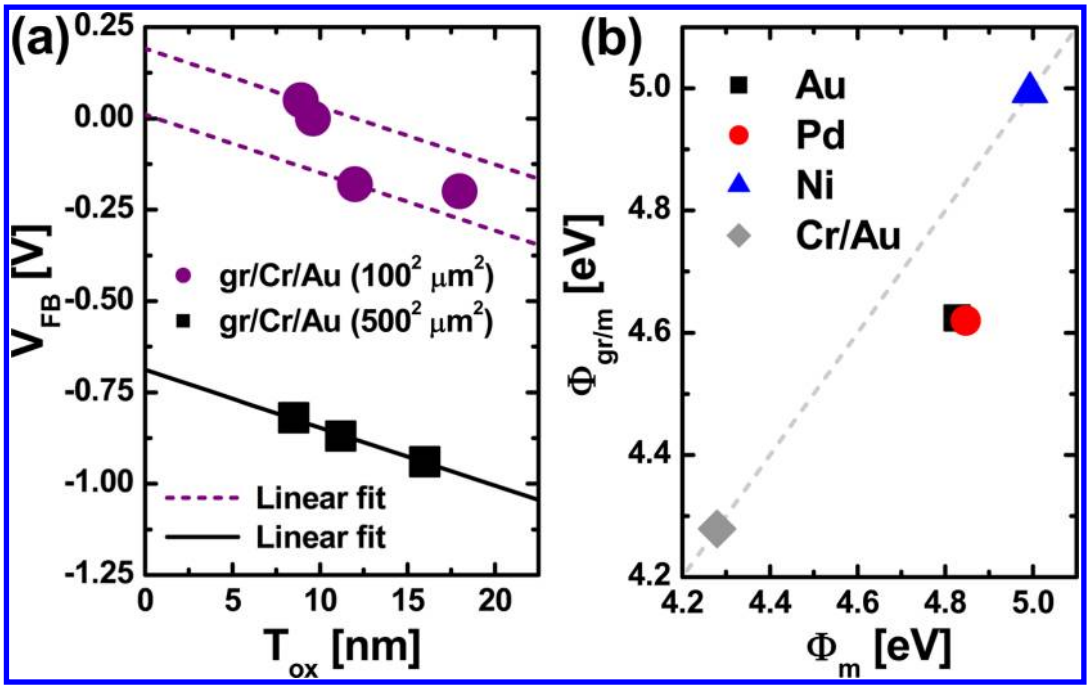

Figure 2. (a) Flat band voltage versus effective oxide thickness (V_{FB} vs EOT) obtained from MGOS capacitors. The solid line is the linear fit of data in the case of full coverage of metal on graphene (metal area = 500 × 500 μm^2), and dashed lines are plotted with the same slope as the solid fit line to extract the work function of exposed graphene. (b) The extracted work function of graphene under the metal ($\Phi_{gr/m}$) plotted against the work function of the corresponding metal.

interesting observation is that as the coverage of the metal pad over graphene increases the $C-V$ curves progressively shift from right to left with noticeable kinks in the depletion region (Figure 1b). Such kinks are often observed in $C-V$ curves when there is a sizable amount (more than 10^{11} cm^{-2}) of interface trapped-charges at the silicon and gate oxide interface.[36] In this case, however, because the silicon and oxide interfaces for all samples are identical, the origin of the kinks in the $C-V$ curves is not interface trapped charges. As the only difference among the samples lies in the gate electrode, it is believed that the kinks are due to the difference in the V_{FB} between graphene and the graphene/metal electrode, and the transition of the $C-V$ curves results from the combination of two V_{FB} values with different portions according to the coverage ratio of metal over graphene.

In order to clarify this, the $C-V$ curves were differentiated according to gate voltage, and a $\partial C/\partial V_g$ versus V_g plot is shown in Figure 1c. Two peaks positioned at ∼−0.82 and ∼0 V, having different heights according to the different coverage ratio of metal over graphene, are clearly observed. For the control sample (no graphene), there is one obvious peak at V_g = −0.82 V, which corresponds to the V_{FB} of the control sample. For the sample with minimal metal coverage (Cr/Au area =100 × 100 μm^2; metal coverage over the graphene electrode is only 4%), there is also one distinct peak at V_g = 0 V; this is considered to be the V_{FB} of the graphene electrode, because the metal area is negligible. As the metal coverage over graphene increases, the peak at V_g = 0 V decreases but the peak at −0.82 V increases. This tendency is more clearly presented in the inset of Figure 1c. Therefore, it is clear that the kink originates from the combined contributions of the capacitance by the metal covered graphene region and the capacitance by the exposed graphene region, and the change of the relative intensities of the peaks is due to the relative magnitudes of the capacitances. This can be modeled as a parallel capacitor of $C_{gr/m}$ (with V_{FB} of the metal contacted graphene electrode) and C_{gr} (with V_{FB} of exposed graphene) with the corresponding area of graphene/metal ($A_{gr/m}$) and bare graphene (A_{gr}), as illustrated in the inset of Figure 1e. In this analysis, one important finding is that the work function of graphene under Cr/Au metal does not correspond with the work function of exposed graphene but rather it follows the work function of the overlying metal.

The same experiment was carried out again but with a different metal electrode, Pd, and the results are shown in Figure 1d. Similar transition behavior of $C-V$ curves, depending on the coverage ratio of the Pd electrode on graphene, is observed as compared to the case of Cr/Au. However, whereas the $\partial C/\partial V_g$ curves of the MGOS capacitors in Figure 1e show two different peaks, the $\partial C/\partial V_g$ peak of the MOS capacitor (pure Pd electrode without graphene) does not match with any peak of the MGOS capacitors. This indicates that the work function of graphene under Pd does not follow the work function of Pd but settled to another value that is even lower than that of Pd. This is different from the case of Cr/Au.

The work function of a gate electrode in a MOS device structure can be obtained by linear extrapolation of a V_{FB} versus EOT (effective oxide thickness) plot, based on the equation below[37]

$$V_{FB} = \Phi_{ms} - \frac{Q_f}{C_{ox}} = \Phi_{ms} - \frac{Q_f T_{ox}}{\varepsilon_{ox}} \quad (1)$$

where $\Phi_{ms} = \Phi_m - \Phi_s$ is the difference in the work function between the gate metal electrode (Φ_m) and a semiconductor (Φ_s), where the latter depends on the doping concentration in silicon. Q_f is the amount of fixed oxide charge at the interface between Si and SiO_2, ε_{ox} is the SiO_2 permittivity, and T_{ox} is the gate oxide thickness (EOT is a general term for various dielectric materials). For this investigation, MGOS devices with

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various oxide thicknesses ranging from 7 to 18 nm were fabricated. The V_{FB} values of graphene/Cr/Au were determined by fitting the experimental $C-V$ curves with simulation results by using a quantum mechanical (QM) CV simulator,[38] as widely used in the silicon device industry. The obtained V_{FB} values are plotted against EOT, as shown in Figure 2a. Φ_{ms} can then be obtained from the intersection of the y-axis. The results also clearly show that the work function of graphene is very different if graphene is covered by a metal. For the case of full coverage (500×500 μm^2 of Cr/Au), the corresponding work function of graphene under Cr/Au metal ($\Phi_{gr/Cr/Au}$) is found to be ~4.3 eV, which is close to the work function of Cr/Au. This agrees well with the results of the $\partial C/\partial V_g$ peak position analysis presented in Figure 1c. The V_{FB} data of exposed graphene (the sample with metal coverage of 100×100 μm^2) do not produce a straight line, as seen in Figure 2a. This is likely due to the inconsistent amount of doping in graphene from sample to sample. However, the slope of the linear fit lines must be identical for all the samples, because the Q_f values of all samples are identical. The plot thus provides an estimated range of the graphene work function, which is 4.89−5.16 eV.[19,20] This range is much higher than the theoretical value (~4.5 eV).[21,22] Previous reports have explained that graphene exposed to air is strongly doped into p-type.[39−42]

We have expanded this analysis to other metals, including nickel, gold, and palladium, which are widely used for graphene FET fabrication, and the results are summarized in the plot of Figure 2b. It has been found that graphene under a Ni electrode shows similar behavior to the case of Cr/Au. The work function of graphene under Ni is simply determined by the work function of Ni (~5 eV). In contrast, the work function of graphene under Pd or Au does not follow the work function of the metal. The results of a $C-V$ analysis for the Au electrode are very similar to those for Pd, which are shown in Figure 1d, and the measured work function of ~4.62 eV for graphene under the Pd contact agrees well with the reported values of the potential step at the graphene/Pd interface (approximately 95−120 meV).[26−28] It is very interesting that the work function of graphene under Pd or Au takes a particular value (~4.62 eV) regardless of the work function of the metal, whereas the work function of graphene under Cr or Ni is pinned to that of the metal electrode. These results suggest that Cr and Ni have stronger interaction with graphene than the other two metals. Because of this strong interaction, when situated under Ni, graphene has been theoretically predicted to lose its unique properties.[15,21,22] However, our $C-V$ measurement results extended to high gate bias (strong accumulation mode) clearly show that the quantum capacitance of graphene is preserved for graphene under a Ni contact (Figure S2, see the details in the Supporting Information). The same tendency is also found in graphene under Pd. This is strong evidence that the characteristic conical points at the K space of graphene under those metals are not destroyed,[43] which conflicts with the previous simulation prediction. Similar discord between experimental and calculation results has also been reported for contacts between carbon nanotubes and metal,[44] and inconsistency among the calculations has been reported[45] as well. Therefore, we believe that the previous theoretical prediction may have some limitations to explain the behavior of the work function of graphene under a metal electrode. Furthermore, the observation that the characteristic conical points at K of graphene under metal are intact also indicates that the work function of graphene is important to determine the nature of the contact between metal and graphene.

Although the exact chemical or physical mechanism underlying this phenomenon for different metals is not clear at present and requires further research, at the very least we anticipate that such different behaviors of the graphene work function under different metal electrodes will affect the contact resistance. In this regard, the gate voltage (V_G) dependent contact resistances for the above four different metal electrodes have been investigated and the results are shown in Figure 3. In

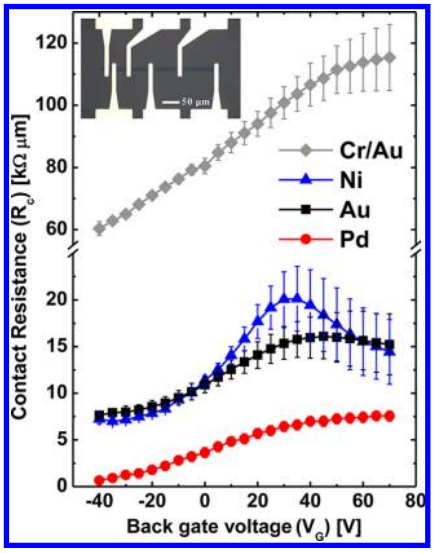

Figure 3. Back gate voltage dependence of width normalized contact resistance (R_c) for different metal contacts. The inset shows the optical microscope image of the fabricated graphene FET arrays with a TLM pattern. The graphene channel width is 10 μm.

this experiment, back gate transistor arrays with a graphene channel width (W) of 10 μm were fabricated on 100 nm thick SiO_2. The width normalized contact resistance (R_c) is carefully measured using the TLM method (inset of Figure 3) with variation of the channel length from 10 to 90 μm in steps of 20 μm. Because of the variation of the Dirac voltage due to nonuniform doping in the graphene channel, the gate voltage causing the maximum R_c is different among the samples. The results also show interesting gate voltage dependence of R_c for different metals. For the case where the work function of graphene has a particular value of 4.62 eV (the cases of Au and Pd), the change of R_c upon variation of the gate voltage is small and the peak R_c is also small. However, for the case where the work function of graphene is pinned to that of the metal (the cases of Cr and Ni), R_c varies strongly according to the change of the gate voltage, and the peak R_c is also high. In particular, the contact resistance of graphene with Cr is 20−100 times higher than that of graphene with Pd throughout the gate voltage range and is similar to the value in a previous report.[1] The results suggest that a low work function of graphene must be avoided for good contact resistance; however, a high work function of graphene does not guarantee good contact resistance, as seen for the case of Ni. Robinson et al. reported that improved contact resistance could be obtained by introducing defects to graphene at the contact area using O_2 plasma and they noted the absence of a correlation between the work function difference and the contact resistance.[46] However, their graphene was quite defective and was overall very different

from the pristine graphene that we are interested in. As such, their work is not closely related to the present study, which focuses on an analysis of the work function of graphene under metal.

The best result in terms of contact resistance is obtained from the case where the work function of graphene is pinned to an intermediate work function value, independent of the metal work function. However, although the Pd and Au contacts result in similar work functions of graphene, Pd has lower contact resistance. This implies that there is another factor that affects the contact resistance. We attribute this to the difference in the wettability of the metal on graphene. A thin Au layer evaporated on a graphite surface easily migrates and aggregates into ramified islands,[47] thus indicating that the wettability of Au on graphene is poor. Pd is, however, widely known to wet carbon networks. It is thought that (1) the change of the work function, which is a result of the charge transfer, and (2) the wettability, which is determined by a force balance between adhesive and cohesive forces from intermolecular interaction, together contribute to the contact resistance. As it provides an intermediate graphene work function of ~4.6 eV and also offers high wettability, Pd forms the best contact with graphene among the metals evaluated in this study.

In summary, we have measured the work function values of graphene under various metal electrodes for the first time using a $C-V$ analysis. Through this analysis, we clarified the four questions noted in the introduction: (1) Graphene under metal is still graphene, as it has unique quantum capacitance properties. (2) However, the original work function of graphene is not preserved when graphene is under metal. It is either pinned to the work function of the metal or pinned to a particular value regardless of the work function of the metal. (3) It is also found that a high work function of graphene does not guarantee low resistance. Graphene under Pd has a midgap work function of ~4.6 eV but provides the lowest contact resistance and the weakest gate voltage dependence. (4) Au has the same property as Pd in terms of modulating the work function of graphene but its contact resistance is higher than that of Pd. This is attributed to the difference in wettability between the metal and graphene. Therefore, the work function is not the only factor determining the contact resistance. These findings provide important information for the realization of high performance graphene−based electronic devices.

■ ASSOCIATED CONTENT

Supporting Information

Description of the extraction method of the work function of silicon substrate, the analysis of quantum capacitance of graphene in MGOS capacitors, and Supplementary Figures S1 and S2 showing Raman spectrum of CVD grown graphene and the quantum capacitance of graphene extracted from $C-V$ curves. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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